

Configuration File (YAML)

engines · PE placement · memory technology · network · power

Workloads (Shared Memory / Message Passing)

μkernels · BabelStream · Rodinia · NPB · SPLASH-3 · PARSEC

Single simulation process — host & device accounting domains (QEMU user-mode + ZSim)

Host Simulation Engine

Core

Core

Core

Core

core_type: ooo / in-order / simple

Caches

L1+L2

Network

GARNET

Memory
Controller

Event Queue

offload region (ROI)

host → device

device → host

H/D transfers are charged on
link + memory models
(PCIe · CXL ·
NVLink-class · interposer)

Device Simulation Engine

PE

PE

PE

PE

pe_type: ooo / in-order / simple / alu / null

placement: subarray ▶ bank ▶ bank-group ▶ chip ▶ rank ▶ channel ▶ logic-die

Address Translation

*TLB + page table
unified / discrete*

In-memory Network

*GARNET
(hierarchical H-tree for DRAM)*

Modeling Infrastructure

Common Interface for External Models (YAML)

DRAM

(Ramulator2)

SRAM

(CACTI)

NVM

(NVSim)

Compute

(ZSim / QEMU)

Network

(GARNET)

Power & Area

(McPAT + memory models)